

Flight Phase: **Takeoff**

Input name	Value	Label	Degree
Turbulence	1.50	Light	0.04
Air speed	125.88	Slow	0.21
Altitude	22424.60	Medium	0.26
Pitch angle	-3.53	Level	0.65
Vertical speed	1202.30	Climb	0.70
Engine health	1.40	Degraded	0.41
Fuel status	35.45	Medium	0.22
Proximity to obstacles	413.71	Caution	0.45
TCA's conflict level	0.40	None	0.41

Output name	Value	Label	Degree
Thrust command	37.60	Low	0.49
Elevator deflection	-15.00	Down	1.00
Autopilot mode suggestion	0.34	Manual	0.53
Stall warning	1.00	Advisory	1.00
Fuel reserve command	1.00	Economy mode	1.00
Stress reduction command	0.34	Normal	0.52

Explanation
- Low airspeed and pitch attitude indicate high stall risk; issuing stall advisory is critical.

Main Contributing Factors
EMPTY

Flight Phase: **Approach**

Input name	Value	Label	Degree
Turbulence	2.89	Severe	0.96
Air speed	93.63	Stall	0.64
Altitude	23.80	Low	1.00
Pitch angle	16.97	Nose up	0.61
Vertical speed	841.39	Climb	0.34
Engine health	0.47	Optimal	0.29
Fuel status	8.94	Empty	0.21
Proximity to obstacles	32.98	Immediate danger	1.00
TCA's conflict level	2.81	Emergency avoidance	0.82

Output name	Value	Label	Degree
Thrust command	100.00	Max	1.00
Elevator deflection	-20.00	Down	1.00
Autopilot mode suggestion	0.00	Manual	1.00
Stall warning	1.00	Warning	1.00
Fuel reserve command	0.00	Emergency diversion	1.00
Stress reduction command	1.00	Override manual	1.00

Explanation
- Low airspeed and pitch attitude indicate high stall risk; issuing stall advisory is critical. - Vertical dynamics suggest aggressive altitude change, requiring corresponding thrust and control. - Engine health degraded; thrust adjusted to ensure performance under failure conditions. - High TCAS conflict level; autopilot mode and manual override may be prioritized for safety.

Main Contributing Factors		
Turbulence	2.89	Severe
Altitude	23.80	Low
Proximity to obstacles	32.98	Immediate danger

FAIL SAFE MODE ACTIVATED

A critical condition was detected
Emergency control outputs were issued.
Fuzzy logic system was overridden for safety response.